

Impact of Imbalanced Data on Model Accuracy

Viola Zeneli, Sidrit Rrustemi, Artemisa Hasalami

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PART 01

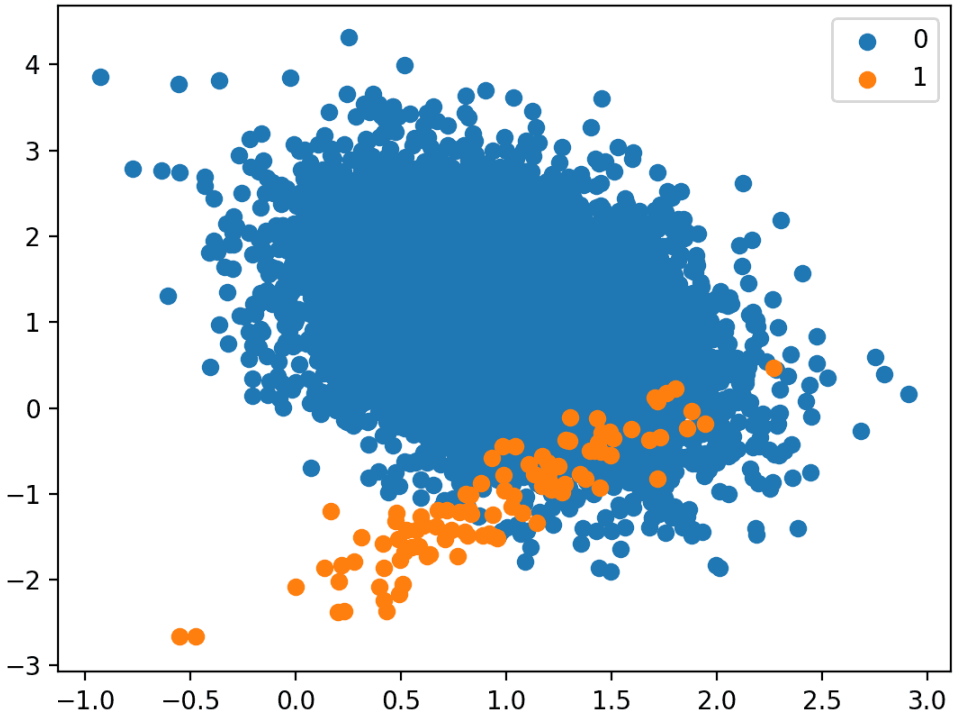
Introduction to Imbalanced Data



What is Imbalanced Data?

Definition of Imbalance

Imbalanced data refers to datasets where the classes are not represented equally; typically one class has a significantly higher number of instances than the others. This can lead to biased models that favor the majority class and struggle to properly learn the characteristics of the minority class. This disparity poses challenges for machine learning models leading to biased outcomes.



Real-World Examples of Imbalanced Data

Healthcare & Rare Diseases

In healthcare, diseases like cancer are fortunately rare, resulting in far fewer positive (cancer) instances than negative (no cancer) instances in datasets. This imbalance makes it challenging for machine learning models to detect these rare cases, even though accurate detection is critical for saving lives.

Fraud & Anomaly Detection

Fraudulent transactions are significantly less frequent than legitimate ones, resulting in an imbalanced dataset. Identifying these rare cases is crucial to prevent financial loss, yet the low frequency of fraud makes it harder for machine learning models to detect them reliably

Customer Churn & Risk Analysis

Only a small percentage of customers may churn, making it important to accurately predict which customers are likely to leave to reduce revenue loss. The low frequency of churn cases can make it harder to predict which customers are at risk.

PART 02

Causes and Origins of Imbalance



Data Collection Issues

01

Sampling Bias

Non-representative sampling can misrepresent class distributions, leading to over- or under-representation of certain classes. This occurs when the collected data does not adequately reflect the full diversity of the population. As a result, machine learning models may become biased, performing well on majority classes but poorly on minority classes. It is important to ensure comprehensive sampling strategies to reduce bias and improve model reliability.

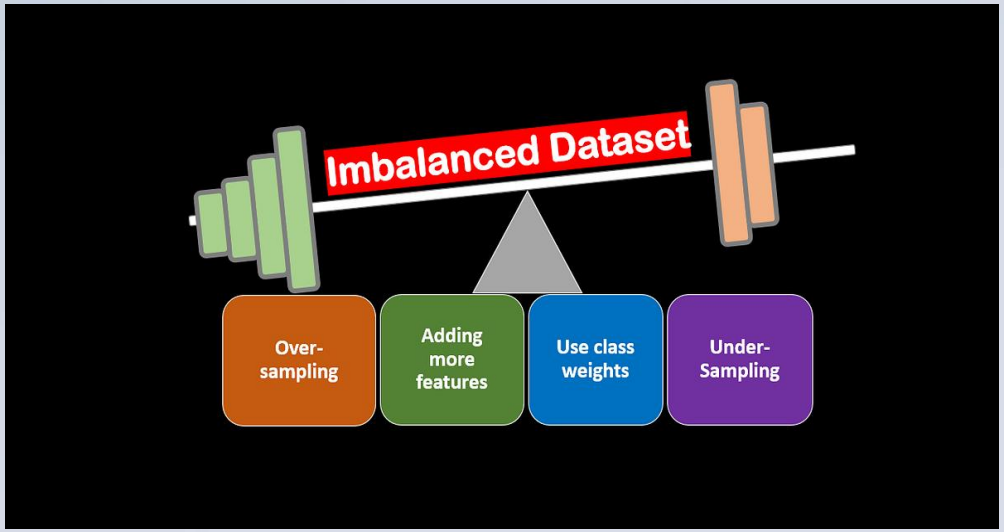
02

Data Scarcity

Limited accessibility to minority class data can worsen imbalance, making it difficult for machine learning models to learn meaningful patterns and generalize effectively. This scarcity may arise due to rare events, high data collection costs, or privacy constraints, necessitating methods such as data augmentation, synthetic data generation, or oversampling to increase minority class representation and improve model performance.

Natural Rarity of Events

Many real-world phenomena involve events that occur infrequently by nature, such as equipment failures, natural disasters, rare diseases, or uncommon genetic mutations. Because these events happen so rarely, collecting a sufficient number of examples is difficult, resulting in highly imbalanced datasets where the minority class is severely underrepresented. This scarcity of minority class samples makes it challenging for machine learning models to learn meaningful patterns, often leading to poor prediction performance for these critical but rare events.



Labeling and Biases

01

Labeling Errors

Incorrect or inconsistent labeling can further distort the distribution of classes and introduce systematic errors in the dataset. Labeling errors may occur due to human mistakes, ambiguous definitions, or inconsistencies in data collection processes. These errors can mislead machine learning models, causing them to learn incorrect patterns and reduce prediction accuracy. Therefore, validating and cleaning labels rigorously is essential to ensure reliable model performance.

02

Preprocessing Bias

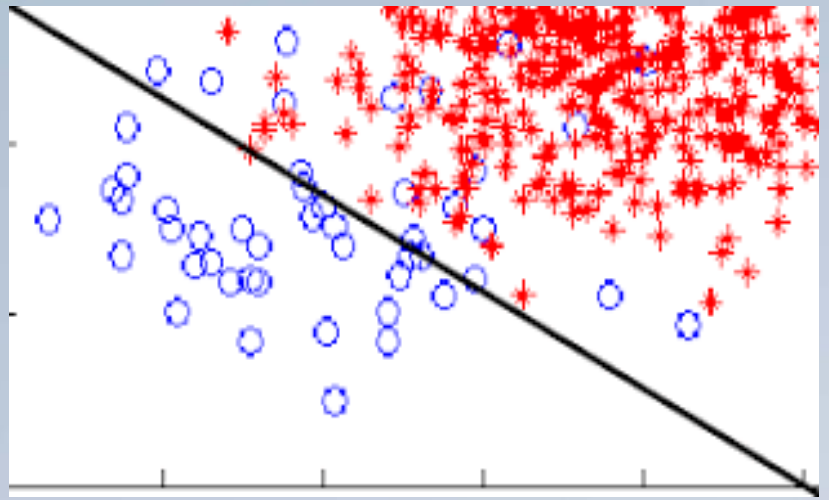
Preprocessing steps, such as feature selection, normalization, or data transformation, can unintentionally amplify existing class imbalances or introduce new biases. For example, removing features important for minority classes or applying transformations unevenly across groups can reduce model accuracy for underrepresented cases. Careful evaluation and testing of preprocessing steps are essential to ensure that models learn fair, accurate, and reliable patterns.

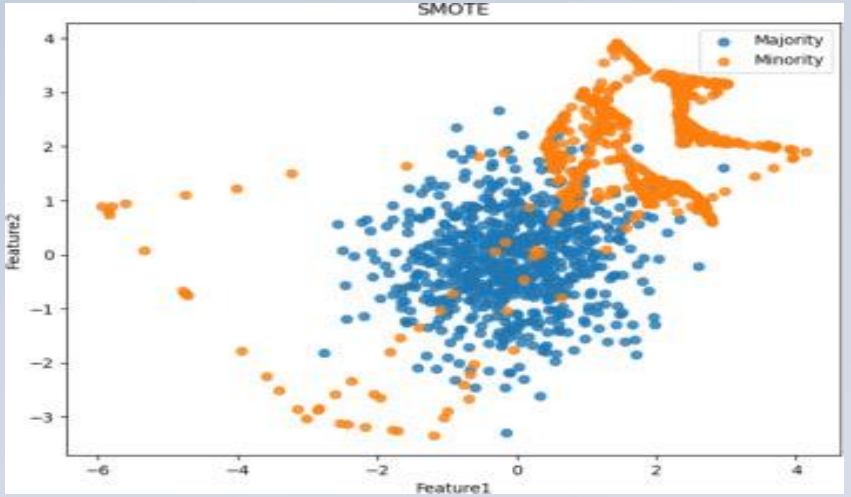
PART 03

Impact on Machine Learning



Bias Toward Majority Class





Skewed Predictions

Skewed predictions occur when models tend to predict the majority class more frequently. This happens because the model is trained on imbalanced data and sees far more examples of the majority class. As a result, the model often performs poorly on minority class predictions, leading to low recall or F1-score for these underrepresented cases. Such bias can reduce the overall effectiveness of the model, especially when correctly predicting minority cases is critical.

Example Scenarios

In credit risk, a model trained on imbalanced data might approve most applicants, even if some are high risk, because approvals dominate the training set. This bias can lead to incorrect predictions, increasing the chance of financial losses, and highlights the need to carefully handle class imbalance during model training.

Poor Minority Class Learning

Difficulty in Pattern Recognition

Patterns in the minority class are often overshadowed by the majority class during training, causing the model to struggle in recognizing important distinguishing features. This can lead to low recall, misclassification of minority instances, and overall reduced model effectiveness. In critical applications, such as fraud detection or medical diagnosis, failing to learn minority patterns can result in serious errors or overlooked cases.

Lack of Sensitivity

The model becomes insensitive to variations in the minority class, making it difficult to accurately identify and classify these events. This lack of sensitivity often results from imbalanced training data, where the majority class dominates learning. Consequently, minority instances may be misclassified, reducing recall and F1-score, and in critical applications like fraud detection or medical diagnosis, this can lead to serious errors or missed cases.

Overfitting and Generalization

01

Overfitting to Majority Class

When trained on imbalanced data, models tend to focus on accurately predicting the majority class, which can lead to overfitting. This means the model learns patterns that work well for the majority class in the training data but fail to generalize to unseen data, especially for the minority class. As a result, the model may perform poorly on new instances, producing inaccurate predictions for underrepresented cases and reducing overall reliability.

02

Generalization Errors

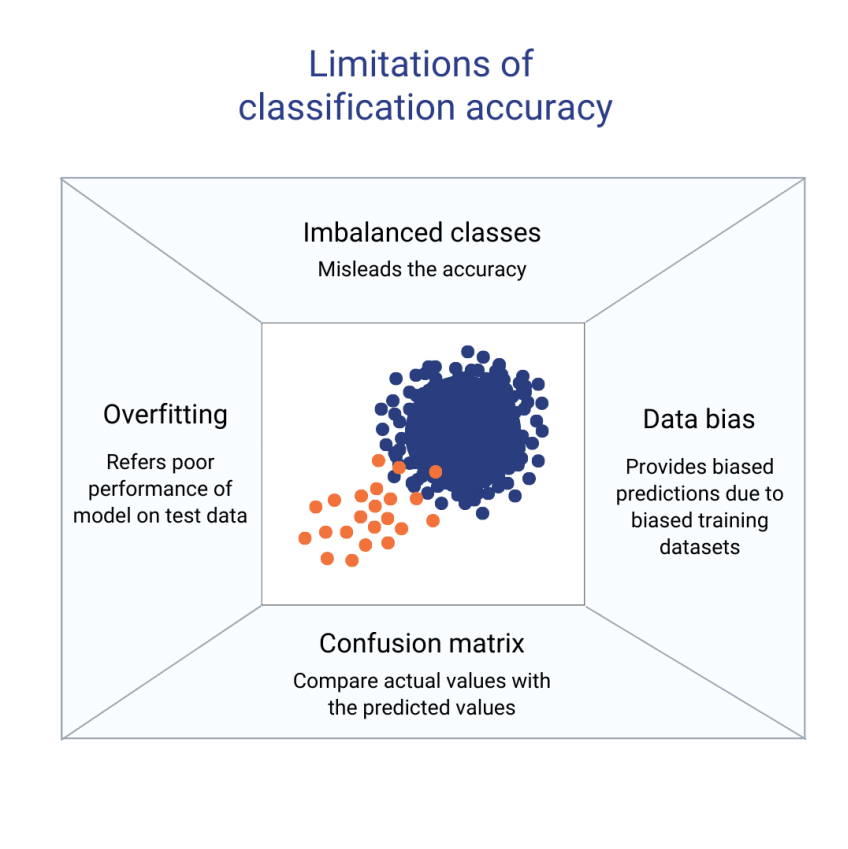
Generalization performance often drops when models fail to capture informative patterns from the minority class. Because the model has been trained primarily on the majority class, it may underperform on unseen data, especially for underrepresented cases. This can lead to misclassifications, low recall or F1-score for the minority class, and reduced reliability in real-world scenarios, such as fraud detection, medical diagnosis, or employee attrition prediction.

PART 04

Model Evaluation Challenges



Limitations of Accuracy



Accuracy can be misleading on imbalanced datasets, as a model may achieve high accuracy by mostly predicting the majority class. Metrics like precision, recall but mainly F1-score, are better for evaluating performance on minority classes.

Precision, Recall and F1 Score



Precision

Precision measures the accuracy of positive predictions and indicates how many predicted positives are actually positive (True Positives / (True Positives + False Positives)).



Recall

Recall measures the ability of the model to find all the positive instances and indicates how many actual positives were correctly classified by the model (True Positives / (True Positives + False Negatives)).



F1 Score

F1 score is the harmonic mean of precision and recall which balances both metrics. It’s useful when there is uneven class distribution.

PART 05

Consequences of Ignoring Imbalance



Healthcare Risks

Misdiagnosis

Failure to detect rare diseases can lead to delayed treatments, misdiagnosis, and severe health consequences.

Incorrect Treatment

Inaccurate diagnosis results in patients receiving treatments that are not tailored to their conditions, resulting in worsened health outcomes.

Financial Loss

Undetected Fraud

Failure to identify fraudulent transactions results in significant financial loss for businesses and individuals.

Regulatory Fines

Businesses may incur regulatory fines for failing to detect and prevent fraudulent activities due to an ineffective model.

Safety and Ethical Concerns

Safety Hazards

In automated systems, failing to detect rare but critical events, such as equipment malfunctions in autonomous vehicles, leads to safety incidents.

Ethical Implications

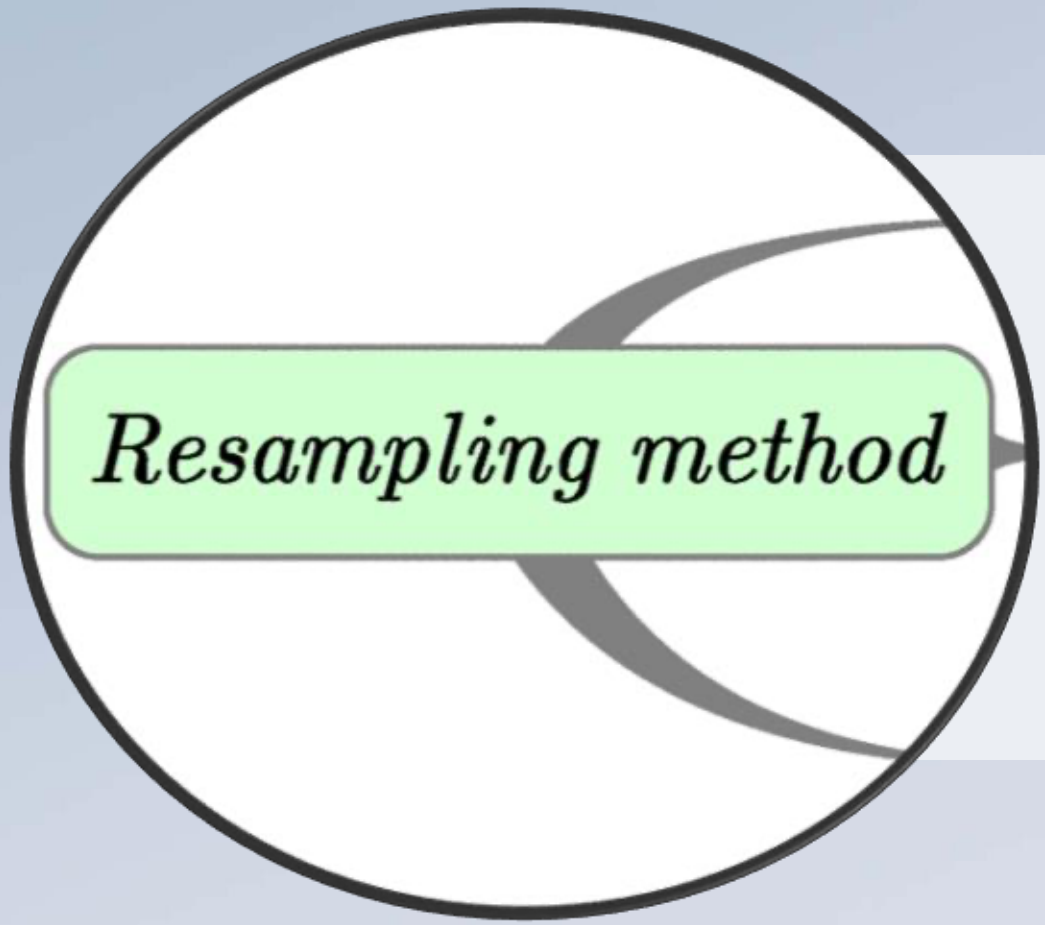
Bias in automated decisions can perpetuate societal inequalities, leading to concerns about fairness and discrimination.

PART 06

Techniques for Handling Imbalanced Data



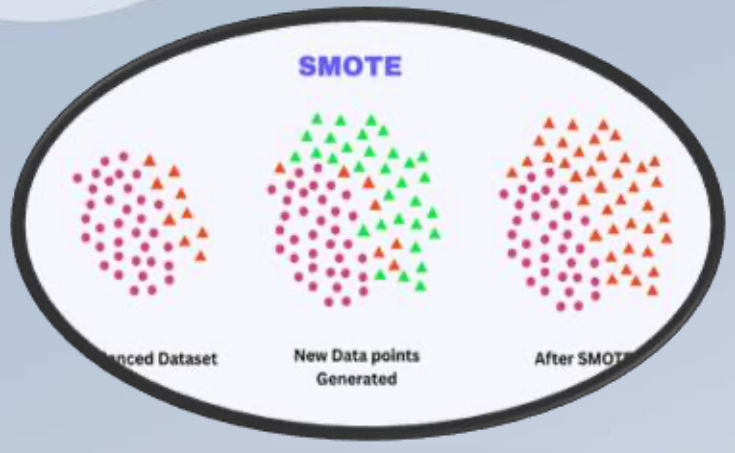
1. Data-Level Approaches



Resampling Methods Overview

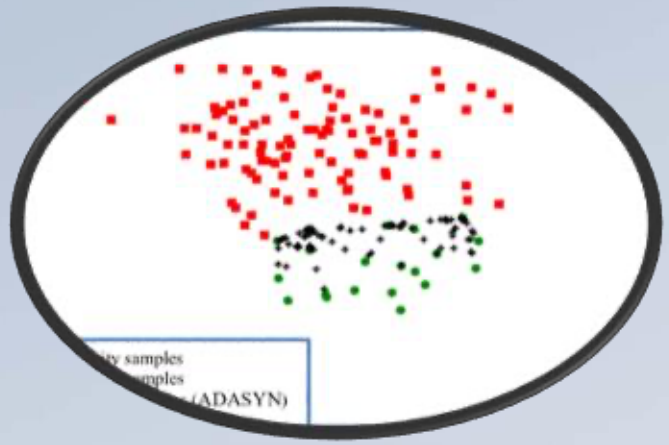
Resampling methods include oversampling the minority class and/or undersampling the majority class to achieve a more balanced dataset. These approaches help mitigate bias toward the majority class and improve the model's ability to learn patterns from underrepresented instances. While effective, care must be taken, as oversampling can lead to overfitting and undersampling may discard useful information from the majority class.

Oversampling Techniques



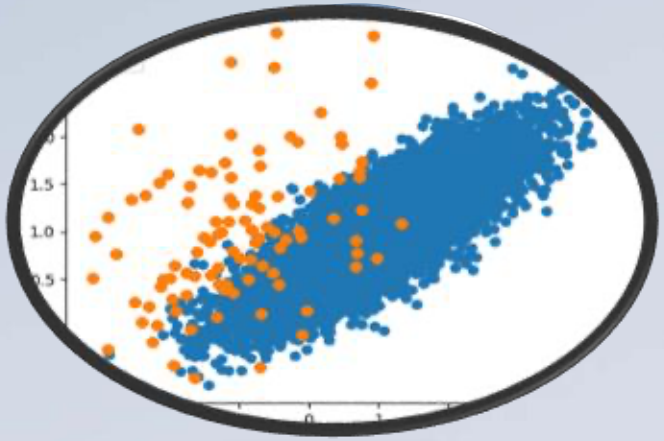
SMOTE

The Synthetic Minority Oversampling Technique (SMOTE) creates new synthetic examples for the minority class by interpolating between existing instances, helping to balance the dataset and improve model performance.



ADASYN

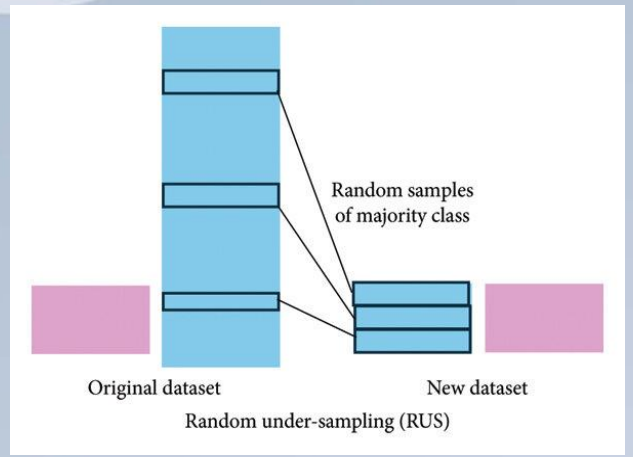
The Adaptive Synthetic Sampling Approach (ADASYN) generates additional synthetic samples for the minority class, focusing on regions where instances are harder to learn, thereby improving the model's ability to capture challenging cases.



Considerations for Oversampling

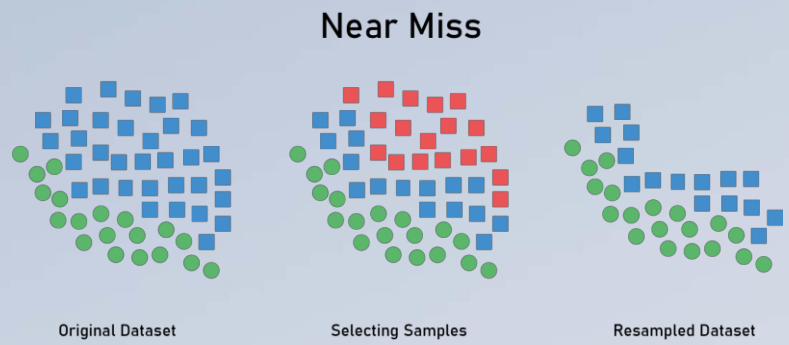
Oversampling can lead to overfitting if not applied carefully, particularly when synthetic samples are very similar to existing minority class instances, causing the model to memorize rather than generalize.

Under-sampling Techniques



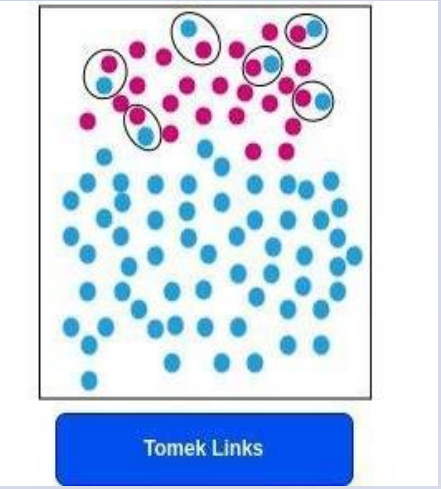
Random Under-sampling

Random under-sampling reduces the majority class by randomly removing instances until the dataset is balanced with the minority class. While simple, this method can result in the loss of potentially valuable information from the majority class, which may affect model performance.



NearMiss

NearMiss selects majority class instances that are closest to minority class instances in the feature space, aiming to retain the most representative majority examples while reducing class imbalance.



Tomek Links

Tomek Links removes pairs of instances from opposite classes that are very close in the feature space, helping to clean class boundaries and reduce overlap between classes.

2. Algorithm-Level Approaches

01

Cost-Sensitive Learning

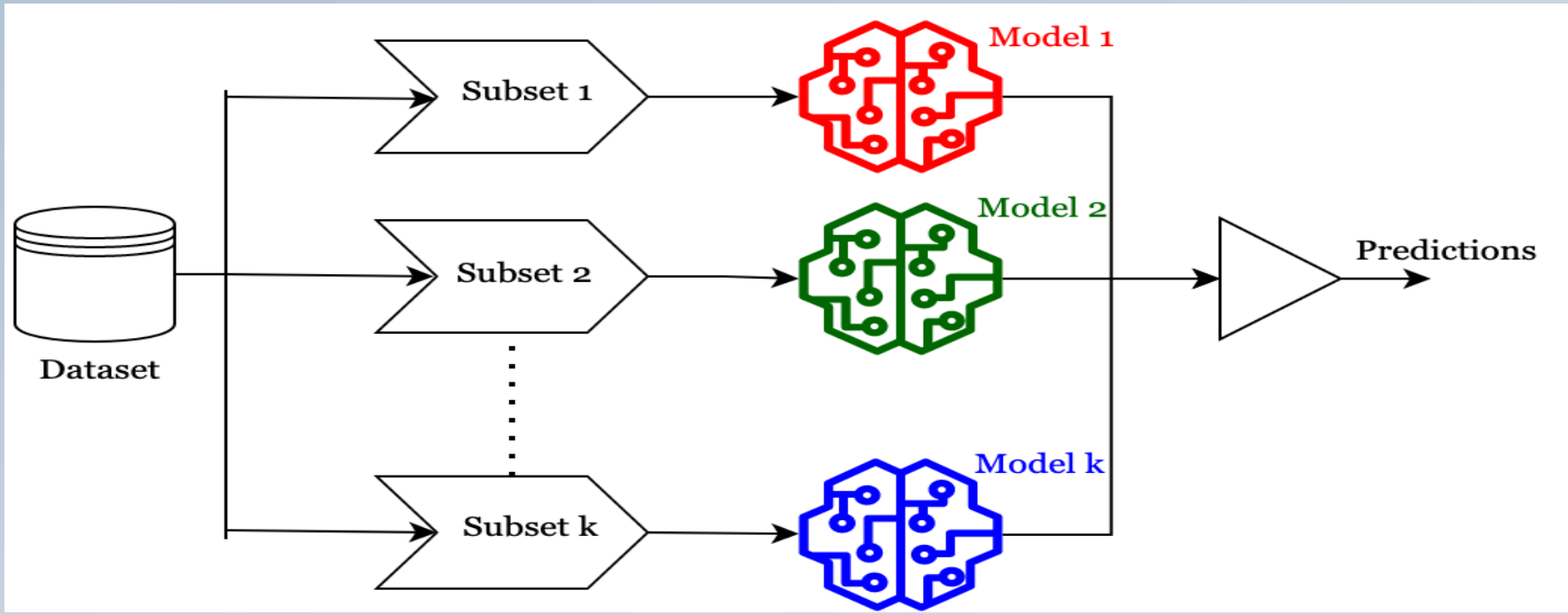
Cost-sensitive learning is a modelling approach that **defines a general framework for assigning different costs to different types of misclassification errors**. Instead of minimizing overall error, the model focuses on minimizing total cost, which is especially important when mistakes involving the minority class have a higher real-world impact. This approach can be implemented through various mechanisms, including modified loss functions or weighting strategies.

02

Adjusting Class Weights

Adjusting class weights is a technique that **provides a specific and practical way to implement cost-sensitive learning** by increasing the penalty for misclassifying minority class samples during training. By forcing the algorithm to pay more attention to underrepresented classes, this method improves minority class recognition without altering the original data distribution.

Ensemble Learning Methods



01 Balanced Random Forests

Balanced Random Forests create balanced bootstrap samples for each decision tree and incorporate class weighting, leveraging ensemble learning to reduce bias toward the majority class and improve overall performance.

02 EasyEnsemble and BalanceCascade

EasyEnsemble and BalanceCascade train multiple learners using different under-sampled subsets of the majority class and then combine their predictions to build a stronger and more robust model.

3. Hybrid Approaches

Hybrid Techniques

Hybrid approaches **combine data-level techniques and algorithm-level techniques** to address class imbalance more effectively than using a single method alone. Instead of only modifying the dataset or only adjusting the learning algorithm, these approaches leverage the strengths of both strategies.

Typically, the dataset is first balanced using methods such as oversampling or under-sampling, and then the model is trained using techniques like class weighting or cost-sensitive learning. **The key advantage of hybrid approaches is that they improve minority class representation while simultaneously guiding the model to focus on important errors**, leading to better overall performance on imbalanced datasets.

Hybrid approaches are especially useful in highly imbalanced problems, where a single technique may be insufficient to capture minority class patterns accurately.

Examples

SMOTE + XGBoost
The dataset is first balanced using SMOTE to increase minority class samples, then XGBoost is trained to learn complex patterns while improving minority class prediction.

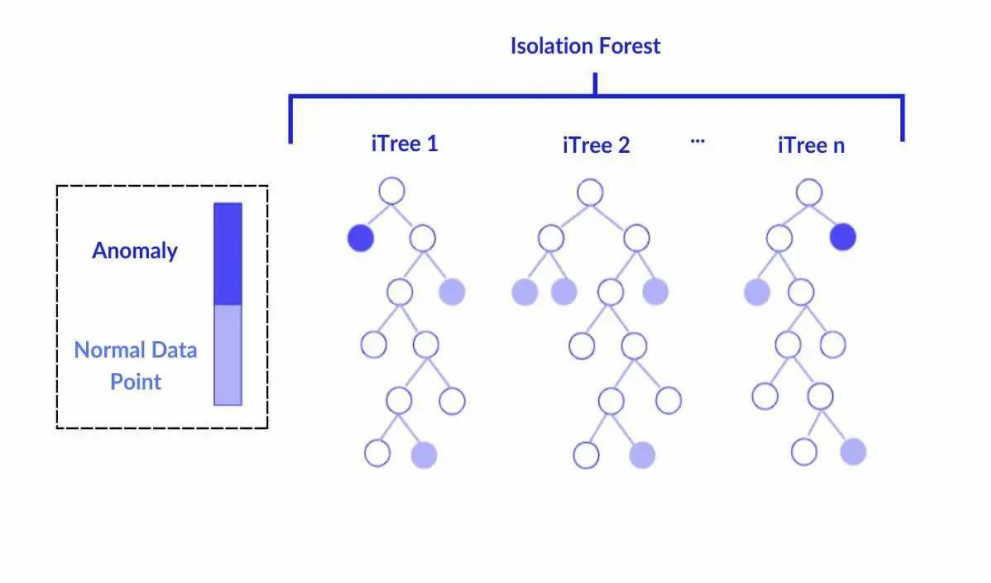
SMOTE + Random Forest
Oversampling improves minority class representation, while Random Forest reduces overfitting and captures non-linear relationships.

SMOTE + Logistic Regression
SMOTE balances the data, allowing logistic regression to learn fairer decision boundaries for the minority class.

Under-sampling + Ensemble Models
Majority class samples are reduced before training ensemble models, improving balance without increasing dataset size.

Class Weighting + Gradient Boosting
Higher penalties are assigned to minority class errors during training, encouraging the model to focus on rare events.

4. Advanced Machine Learning Techniques



Anomaly Detection Algorithms

Anomaly detection techniques, such as **Isolation Forest** and **One-Class SVM**, are mainly used when the dataset is **extremely imbalanced** or when **very few labelled minority samples are available**. Instead of learning both classes explicitly, these methods model the normal (majority) behaviour and identify deviations from it as anomalies. This approach can be more effective than standard supervised classification when minority class examples are too scarce to learn reliable patterns. Anomaly detection is particularly useful in scenarios such as fraud detection, network intrusion detection, and rare event monitoring, where abnormal behaviour is well-defined but rare.

PART 07

Employee Attrition Case Study (Our Mini Project)



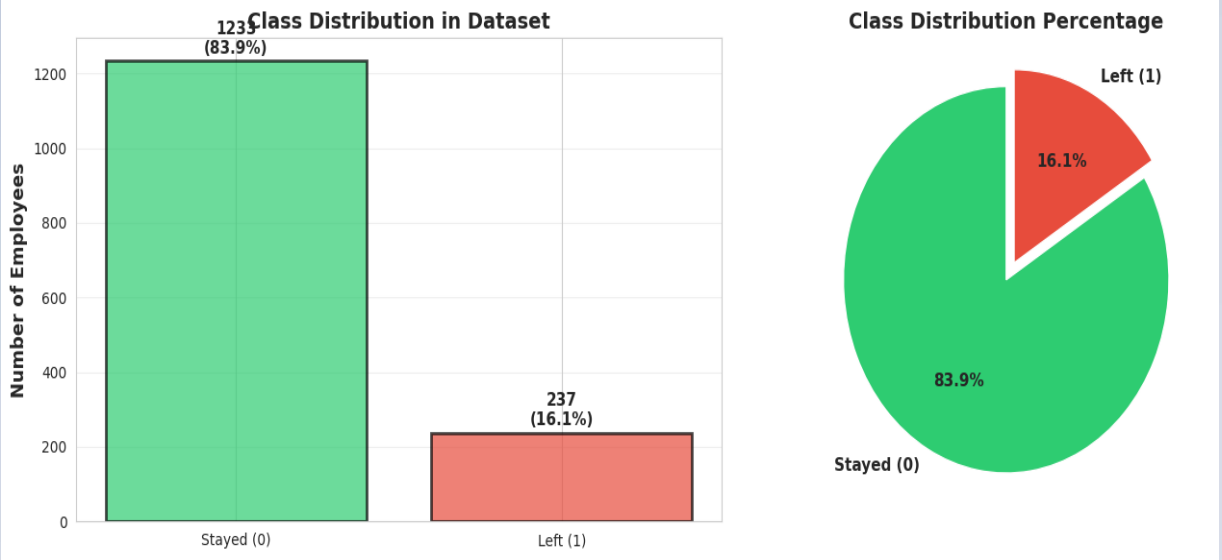
Employee Attrition Case Study

Goals

- Predict whether an employee will leave the company.
- Demonstrate how class imbalance affects model performance and how balancing improves results.

Dataset

- Employee Attrition & Retention Analytics Dataset: https://www.kaggle.com/datasets/ajinkyachintawar/employee-attrition-and-retention-analytics-dataset
- Features: Age, Salary, Job Satisfaction, Work-Life Balance, etc.
- Imbalance: Stayed = 1233, Left = 237



Approach

- Model 1 – Imbalanced Data: Train Random Forest on original dataset.
- Model 2 – Balanced Data: Apply SMOTE-Tomek to balance classes, then train Random Forest.

Key observation:

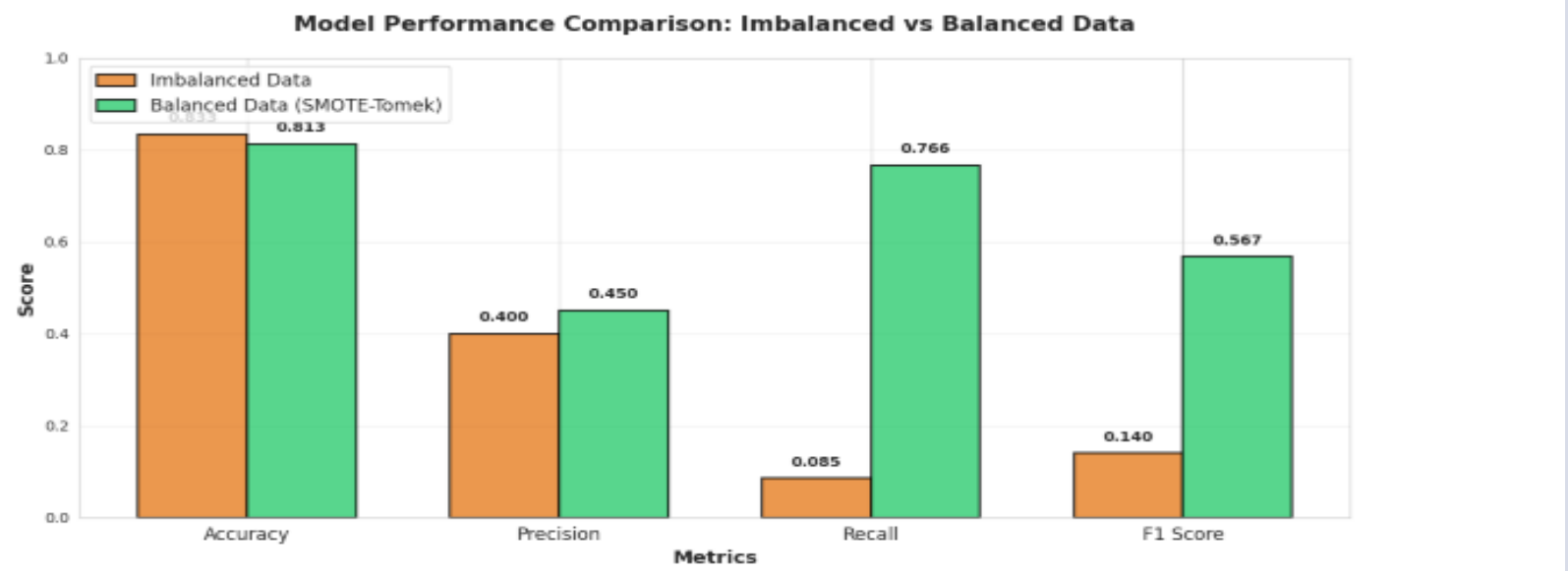
- Imbalanced model misses most employees who leave (Recall: 0.085)
- Balanced model identifies significantly more at-risk employees (Recall: 0.766)
- F1 score improves from 0.14 → 0.57 for minority class

Business Impact:

- Better identification of at-risk employees
- Enables proactive HR interventions and retention strategies

Comparison of Results before & after SMOTE-Tomek:

Metric	Imbalanced Data	Balanced Data (SMOTE-Tomek)	Improvement
Accuracy	0.833	0.813	-2.4%
Precision (Left)	0.400	0.45	+12.5%
Recall (Left)	0.085	0.7659	+800%
F1 Score (Left)	0.14	0.567	+304%



PART 08

Final Conclusions



Final Conclusions

Effective Strategies

Handling class imbalance is essential for achieving reliable and meaningful model performance, as standard accuracy alone can be misleading. The use of **resampling techniques**, **cost-sensitive learning**, and **appropriate evaluation metrics** significantly improves the model's ability to detect minority class instances. Proper validation strategies, such as stratified sampling, further ensure that model performance generalizes well to real-world imbalanced scenarios.

Future Directions

Future work can focus on **hybrid and advanced learning approaches** to better address severe class imbalance. Potential directions include **ensemble-based hybrid models**, **adaptive resampling techniques**, and **cost-sensitive loss functions in deep learning models**. Additionally, developing more **imbalance-aware evaluation strategies** can lead to more robust and accurate decision-making in complex, real-world datasets.



THANK YOU !